# Piecewise-Deterministic Markov Processes

## Theory and Methods for Stationary Distributions

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## 1. Motivation

Davis (1984) observed that continuous-time stochastic models consist of three ingredients: diffusion (Brownian noise), deterministic motion, and random jumps. While diffusion theory is highly unified via Itô calculus, non-diffusion models — combining deterministic motion and random jumps — were a heterogeneous collection of special cases. The goal of PDMPs is to provide a canonical class covering virtually all such models, with a unified stochastic calculus.

**Classical examples:** countable-state Markov processes, M/G/1 and GI/G/1 queues (virtual waiting time), dam and storage models.

**Modern applications:** TCP/IP congestion control, reliability and maintenance models, biochemical reaction networks, insurance ruin theory.

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## 2. Definition of the PDMP

A PDMP on an open set $E \subseteq \mathbb{R}^d$ is determined by three local characteristics:

**(1) The flow $\varphi$.** Unique solution map of $\dot{x} = f(x)$, where $f$ is Lipschitz continuous:

$$\varphi(0, x) = x, \qquad \varphi(t + s, x) = \varphi(t, \varphi(s, x)).$$

**(2) The jump rate $\lambda$.** A measurable function $\lambda : E \to \mathbb{R}_+$. The **integrated hazard along the trajectory** is:

$$\Lambda(t, x) = \int_0^t \lambda(\varphi(s, x))\, ds.$$

This is the cumulative intensity seen by a particle starting at $x$ and following the ODE flow for time $t$. The first jump time satisfies:

$$P(T_1 > t \mid X_0 = x) = e^{-\Lambda(t,x)}.$$

**(3) The jump kernel $Q$.** A probability measure $Q(x, \cdot)$ on $E$. Post-jump locations $Z_i \sim Q(\varphi(T_i, x_0), \cdot)$ are drawn independently of the jump times.

### Reachable boundary and Davis's assumption

The **reachable boundary** is $\partial^* E = \{z \in \partial E : \varphi(-t, z) \in E \text{ for all } t \in (0, \varepsilon)\}$ — those boundary points the flow can reach from the interior. The **boundary exit time** is $t^*(x) = \inf\{t > 0 : \varphi(t, x) \in \partial^* E\}$.

> **Davis's assumption.** If $t^*(x) < \infty$, then $\Lambda(t^*(x), x) = \infty$. Consequence: $P(T_1 < t^*(x)) = 1$, so the Poisson clock always fires before the boundary is reached. Forced boundary jumps are excluded. *Costa (1990) relaxes this assumption.*

### Sample path construction

Starting from $x_0 \in E$, one inter-jump cycle is constructed as follows:

1. **Draw the alarm level.** Sample $U \sim \text{Uniform}(0,1)$ and set $s^* = -\log U$.
2. **Run the flow and clock jointly.** Integrate $\dot{x} = f(x)$ from $x_0$ while accumulating $\Lambda(t, x_0)$. Stop at $T_1 = \inf\{t > 0 : \Lambda(t, x_0) \geq s^*\}$.
3. **Draw the post-jump location.** Sample $Z_1 \sim Q(\varphi(T_1, x_0), \cdot)$ independently of $T_1$.
4. **Restart** from $X_{T_1} = Z_1$ and repeat.

   **Key point.** Steps 1 and 2 cannot be separated: $T_1$ is the *output* of the joint integration, not an input to the evolution.

### Markov property and non-explosion

The semigroup property of $\varphi$ ensures the process is **strong Markov**. Non-explosion requires $E[N_t] < \infty$ for all $t$, equivalently $T_n \to \infty$ a.s. This must be verified for each model.

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## 3. Example: TCP/IP Congestion Control

The TCP/IP AIMD model (additive increase, multiplicative decrease) is a natural PDMP.

| Object | Value |
|---|---|
| State space | $E = \mathbb{R}_+$, $x$ = sending rate |
| Flow | $f(x) = c > 0$, so $\varphi(t, x) = x + ct$ |
| Jump rate | $\lambda(x) = \alpha x$ |
| Jump kernel | $Q(x, \cdot) = \delta_{x/2}$ |

No reachable boundary ($t^*(x) = \infty$ for all $x$), so Davis's assumption holds trivially.

The generator for test functions $\psi \in \mathcal{D}(\mathcal{A})$ is:

$$\mathcal{A}\psi(x) = c\,\psi'(x) + \alpha x \left[\psi\left(\tfrac{x}{2}\right) - \psi(x)\right].$$

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## 4. The Extended Generator

**Theorem (Davis 1984, Theorem 5.5).** For $\psi \in \mathcal{D}(\mathcal{A})$, the extended generator of the PDMP is:

$$\mathcal{A}\psi(x) = \underbrace{f(x) \cdot \nabla\psi(x)}_{\text{ODE drift}} + \underbrace{\lambda(x) \int_E [\psi(y) - \psi(x)]\, Q(x, dy)}_{\text{average jump contribution}}.$$

The domain $\mathcal{D}(\mathcal{A})$ consists of functions $\psi$ satisfying: 1. $t \mapsto \psi(\varphi(t, x))$ is absolutely continuous on $[0, t^*(x))$ for each $x \in E$. 2. **Boundary condition:** $\psi(x) = \int_E \psi(y)\, Q(x, dy)$ for $x \in \partial^* E$ (vacuous under Davis's assumption). 3. **Integrability:** $\int_E [\psi(y) - \psi(x)]\, Q(x, dy) \in L^{1,\text{loc}}(p)$.

### Dynkin formula

For $\psi \in \mathcal{D}(\mathcal{A})$, the process

$$C_t^\psi = \psi(X_t) - \psi(X_0) - \int_0^t \mathcal{A}\psi(X_s)\, ds$$

is a local $\mathcal{F}_t$-martingale. Note that $\mathcal{A}$ is **not a local operator** — the integral term couples $\psi(x)$ to all states reachable by a jump. Unlike diffusion, the calculus here is ordinary calculus with a stochastic interpretation — no Itô correction terms.

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## 5. Kolmogorov Forward Equation

Let $p(x,t)$ denote the density of $X_t$. Taking the $L^2$-adjoint of $\mathcal{A}$:

$$\partial_t p(x,t) = \underbrace{-\nabla \cdot [f(x)\, p(x,t)]}_{\text{transport}} \underbrace{- \lambda(x)\, p(x,t)}_{\text{loss via jumps}} + \underbrace{\int_E \lambda(y)\, p(y,t)\, Q(y,dx)}_{\text{gain via jumps}}.$$

This is a **first-order hyperbolic PDE** (not Fokker-Planck — there is no diffusion term) coupled to a non-local integral operator. The gain term couples the density at all states $y$ jumping into $x$.

**Stationary condition** (set $\partial_t p = 0$): a density $\pi(x)$ is stationary if and only if:

$$\nabla \cdot [f(x)\, \pi(x)] + \lambda(x)\, \pi(x) = \int_E \lambda(y)\, \pi(y)\, Q(y,dx).$$

This is an integro-differential equation, generally hard to solve analytically.

### TCP/IP stationary equation

With $f(x) = c$, $\lambda(x) = \alpha x$, $Q(x,\cdot) = \delta_{x/2}$, changing variables $y = 2x$ in the gain term:

$$c\, \pi'(x) + \alpha x\, \pi(x) = 4\alpha x\, \pi(2x).$$

This couples $\pi(x)$ at scale $x$ to $\pi(2x)$ at scale $2x$ — a **functional differential equation**. Standard ODE methods do not apply directly. The natural tool is the **Mellin transform** (Baccelli & McDonald, 2006).

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## 6. Ergodicity

**Definition.** A PDMP with semigroup $(P_t)$ is **ergodic** if it admits a unique invariant probability measure $\pi$ satisfying $\int_E P_t\psi\, d\pi = \int_E \psi\, d\pi$ for all bounded measurable $\psi$ and $t \geq 0$, and $\mu_t \to \pi$ weakly for all initial conditions $x \in E$.

**Ergodic theorem.** Under ergodicity, for all $x \in E$ and $g \in L^1(\pi)$:

$$\frac{1}{T}\int_0^T g(X_t)\, dt \xrightarrow{T\to\infty} \int_E g\, d\pi \quad \text{a.s.}$$

This justifies Monte Carlo estimation of $\pi$ from a single trajectory.

**Stronger notions:** - **Positive recurrence:** $E_x[\tau_C] < \infty$ for all $x$ and some petite set $C$. - **Geometric ergodicity:** $\|\mu_t - \pi\|_V \leq C_0\, V(x_0)\, e^{-\varrho t}$ for some $V \geq 1$.

### Conditions for ergodicity

**Condition 1: $\nu$-irreducibility** (Meyn & Tweedie 1993, Ch. 4). The PDMP is $\nu$-irreducible with respect to a $\sigma$-finite measure $\nu$ if $\int_0^\infty P_t(x, A)\, dt > 0$ for all $x \in E$ and $\nu(A) > 0$. For a PDMP this requires: (i) ODE reachability — from $x$, the flow must reach a neighborhood of some $y$ from which a jump into $A$ is possible; and (ii) jump support — $Q(y, A) > 0$ for some such $y$.

> **Warning.** If the ODE has an invariant set $\mathcal{L}$ and $Q(x, \mathcal{L}) = 1$ for all $x \in \mathcal{L}$, the process is trapped and irreducibility fails. Irreducibility requires jumps to escape invariant sets of the flow.

Irreducibility is **necessary but not sufficient** for ergodicity.

**Condition 2: Harris recurrence** (Meyn & Tweedie 1993, Ch. 9). A $\nu$-irreducible PDMP is **Harris recurrent** if for every $A$ with $\nu(A) > 0$:

$$P_x\left(\int_0^\infty \mathbf{1}_{X_t \in A}\, dt = \infty\right) = 1 \quad \text{for all } x \in E.$$

Harris recurrence ensures every positive-measure set is visited infinitely often from every starting point.

**Condition 3: Petite sets** (Meyn & Tweedie 1993, Ch. 5). A set $C \subset E$ is **petite** if there exist $T > 0$, $\varepsilon > 0$ and a probability measure $\nu_C$ such that $P^T(x, \cdot) \geq \varepsilon\, \nu_C(\cdot)$ for all $x \in C$.

Every visit to $C$ carries a probability $\varepsilon$ of *refreshing* the distribution to $\nu_C$, breaking dependence on the initial condition. Without petite sets, two trajectories could visit $C$ infinitely often yet never couple. A compact set $C$ is petite for a PDMP if: $\lambda(x) \geq \lambda_{\min} > 0$ on $C$; $Q(x, \cdot)$ has a density bounded below on $C$; and the ODE is locally controllable within $C$.

**Condition 4: Positive recurrence** (Meyn & Tweedie 1993, Theorem 10.0.1). A Harris recurrent PDMP is **positively recurrent** if for some petite set $C$ with $\nu(C) > 0$: $E_x[\tau_C] < \infty$ for all $x \in E$, where $\tau_C = \inf\{t > 0 : X_t \in C\}$. If this holds for one petite set it holds for all. Moreover:

$$\pi(A) \propto E_x\left[\int_0^{\tau_C} \mathbf{1}_{X_t \in A}\, dt\right],$$

i.e. $\pi$ is proportional to the expected time spent in $A$ during one excursion from $C$.

> **Summary.** $\nu$-irreducibility + Harris recurrence + positive recurrence $\Rightarrow$ ergodicity.

### Sufficient conditions

**Foster's criterion** (Meyn & Tweedie 1993, Theorem 3.2; Costa & Dufour 1999, Theorem 3.1). Suppose the PDMP is $\nu$-irreducible. If there exist $V \geq 1$, a petite set $C$, and constants $\varepsilon > 0$, $b < \infty$ such that:

$$\mathcal{A}V(x) \leq -\varepsilon + b\, \mathbf{1}_C(x),$$

then the PDMP is Harris recurrent, positively recurrent, and admits a unique invariant distribution $\pi$. Applying the Dynkin formula and optional stopping:

$$E_x[\tau_C] \leq \frac{V(x) - 1 + b}{\varepsilon} < \infty.$$

**Lyapunov criterion** (Meyn & Tweedie 1993; Durmus, Guillin & Monmarché 2021). If instead $\mathcal{A}V(x) \leq -c\, V(x) + b\, \mathbf{1}_C(x)$ for some $c > 0$, then the PDMP is geometrically ergodic.

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## 7. TCP/IP: Ergodicity Analysis

**Irreducibility.** The TCP/IP PDMP is $\nu$-irreducible with respect to Lebesgue measure. From any $x > 0$, the flow reaches any $y > x$ in time $(y - x)/c$; a jump then sends the process to $y/2$. By choosing $y = 2z$ for any target $z$, any interval $(a, b)$ is reachable in one flow segment plus one jump.

**Petite set.** For any $0 < a < b < \infty$, the compact set $C = [a, b]$ is petite. Since $Q(x, \cdot) = \delta_{x/2}$ is a point mass, one jump alone gives no absolutely continuous component. However, after the first jump the process lands at $x/2$ and flows for a random time before the second jump. Since the flow time varies continuously, the second post-jump location has an absolutely continuous distribution, giving the required minorization.

**Foster's criterion with $V(x) = x$.**

$$\mathcal{A}V(x) = c \cdot 1 + \alpha x \left[\frac{x}{2} - x\right] = c - \frac{\alpha x^2}{2}.$$

Choose $\delta > 0$ small and $C = [\delta,\ \sqrt{2c/\alpha} + \delta]$. For $x > \sqrt{2c/\alpha} + \delta$:

$$\mathcal{A}V(x) \le c - \frac{\alpha}{2}\left(\sqrt{\frac{2c}{\alpha}} + \delta\right)^2 = -\alpha\delta\sqrt{\frac{c}{2\alpha}} - \frac{\alpha\delta^2}{2} =: -\varepsilon < 0.$$

Inside $C$: $\mathcal{A}V(x) \le c =: b$. Foster's criterion holds, giving Harris recurrence, positive recurrence, and a unique stationary distribution $\pi$.

**Second moment from the generator.** Since $\pi$ is stationary, $\int \mathcal{A}\psi \, d\pi = 0$ for all $\psi \in \mathcal{D}(\mathcal{A})$. Choosing $\psi(x) = x$:

$$c - \frac{\alpha}{2} E_\pi[X^2] = 0 \implies \boxed{E_\pi[X^2] = \frac{2c}{\alpha}}.$$

**Bound on the mean.** By Jensen's inequality ($x^2$ is convex):

$$E_\pi[X]^2 \le E_\pi[X^2] = \frac{2c}{\alpha} \implies E_\pi[X] \le \sqrt{\frac{2c}{\alpha}}.$$

The exact value of $E_\pi[X]$ is derived via the Mellin transform in the appendix below.

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## Appendix: Mellin Transform Analysis of the TCP/IP Stationary Distribution

### A.1 The balance equation

The stationary forward equation for the TCP/IP model is:

$$c\,\pi'(x) + \alpha x\,\pi(x) = 4\alpha x\,\pi(2x).$$

This is a **functional differential equation**: it couples $\pi(x)$ at scale $x$ to $\pi(2x)$ at scale $2x$. Standard ODE methods do not apply. The Mellin transform converts this scale coupling into an algebraic recurrence.

### A.2 The Mellin transform

The **Mellin transform** of $\pi$ is:

$$\hat{\pi}(u) = \int_0^\infty x^{u-1}\pi(x)\,dx, \quad u \in \mathbb{C},\ \alpha < \mathrm{Re}(u) < \beta.$$

Since $\hat{\pi}(u) = E_\pi[X^{u-1}]$, integer values give moments: $\hat{\pi}(1) = 1$, $\hat{\pi}(2) = E_\pi[X]$, $\hat{\pi}(3) = E_\pi[X^2]$, etc.

**Key properties:** 1. $\mathcal{M}[\pi'](u) = -(u-1)\hat{\pi}(u-1)$ — integration by parts 2. $\mathcal{M}[x\pi(x)](u) = \hat{\pi}(u+1)$ 3. $\mathcal{M}[\pi(ax)](u) = a^{-u}\hat{\pi}(u)$ for $a > 0$ — substitute $y = ax$

### A.3 The functional equation for $\hat{\pi}$

Applying the Mellin transform to the balance equation using properties 1, 2, 3:

$$-c(u-1)\hat{\pi}(u-1) + \alpha(1-2^{1-u})\hat{\pi}(u+1) = 0$$

which gives the **functional equation**:

$$\boxed{\hat{\pi}(u+1) = \frac{c(u-1)}{\alpha(1-2^{1-u})}\,\hat{\pi}(u-1)}$$

a recurrence in steps of 2, valid for all $u$ in the fundamental strip.

### A.4 The ansatz (Baccelli & McDonald 2006)

Write:

$$\hat{\pi}(u) = g(u)\,\Gamma\!\left(\frac{u}{2}\right)\left(\frac{2c}{\alpha}\right)^{u/2}$$

Substituting into the functional equation, the $\Gamma$ and power factors cancel (using $\Gamma((u+1)/2) = \frac{u-1}{2}\Gamma((u-1)/2)$), leaving the simple recurrence:

$$g(u) = g(u+2)(1-2^{-u})$$

Assuming $g(u) \to g(\infty)$ as $\mathrm{Re}(u) \to \infty$ (proved in Baccelli et al. 2005), iterate forward:

$$g(u) = g(\infty)\prod_{k=0}^{\infty}(1-2^{-u-2k}) =: g(\infty)\,\Pi_\infty(u)$$

where $\Pi_\infty(u) = \prod_{k=0}^{\infty}(1-2^{-u-2k})$.

### A.5 The solution

Substituting back:

$$\hat{\pi}(u) = C\,\Gamma\!\left(\frac{u}{2}\right)\left(\frac{2c}{\alpha}\right)^{u/2}\Pi_\infty(u)$$

**Verification.** The ratio $\Pi_\infty(u+1)/\Pi_\infty(u-1) = 1/(1-2^{1-u})$ follows from extracting the $k=0$ term: $\Pi_\infty(u-1) = (1-2^{1-u})\Pi_\infty(u+1)$. Combined with the Gamma ratio $(u-1)/2$:

$$\frac{\hat{\pi}(u+1)}{\hat{\pi}(u-1)} = \frac{2c}{\alpha}\cdot\frac{u-1}{2}\cdot\frac{1}{1-2^{1-u}} = \frac{c(u-1)}{\alpha(1-2^{1-u})}\checkmark$$

**Normalization.** Setting $\hat{\pi}(1) = 1$ and using $\Gamma(1/2) = \sqrt{\pi}$:

$$C = \frac{1}{\sqrt{2\pi c/\alpha} \cdot \Pi_\infty(1)}$$

### A.6 Second moment: verification

Computing $\hat{\pi}(3) = E_\pi[X^2]$ using $\Gamma(3/2) = \sqrt{\pi}/2$ and $\Pi_\infty(3)/\Pi_\infty(1) = 2$ (from extracting the $k = 0$ term of $\Pi_\infty(1)$):

$$E_\pi[X^2] = \hat{\pi}(3) = \frac{2c}{\alpha} \checkmark$$

Consistent with the generator argument.

### A.7 Mean throughput

Computing $\hat{\pi}(2) = E_\pi[X]$ using $\Gamma(1) = 1$:

$$E_\pi[X] = \sqrt{\frac{2c}{\alpha\pi}} \cdot \frac{\Pi_\infty(2)}{\Pi_\infty(1)}$$

Numerically:

$$\frac{\Pi_\infty(2)}{\Pi_\infty(1)} = \frac{\prod_{k=0}^{\infty}(1 - 4^{-k-1})}{\prod_{k=0}^{\infty}(1 - 2^{-1-2k})} \approx \frac{0.6894}{0.4278} \approx 1.612$$

giving:

$$\boxed{E_\pi[X] \approx 0.91\sqrt{\frac{2c}{\alpha}}}$$

consistent with the Jensen upper bound $E_\pi[X] \leq \sqrt{2c/\alpha}$.

### A.8 Connection to the Mathis formula

For a persistent TCP flow with packet loss probability $p$ and round trip time $R$, the Mathis formula gives $M = K/(R\sqrt{p})$ with $K = \sqrt{3/2}$ for periodic losses. In our model $c = 1/R^2$ and $\alpha = p$, so our formula for random losses gives:

$$E_\pi[X] = \frac{1}{R\sqrt{p}} \cdot \sqrt{\frac{2}{\pi}} \cdot \frac{\Pi_\infty(2)}{\Pi_\infty(1)} \approx \frac{1.286}{R\sqrt{p}}$$

which has the same $1/(R\sqrt{p})$ structure as the Mathis formula, with the constant determined by the stochastic loss process.

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